

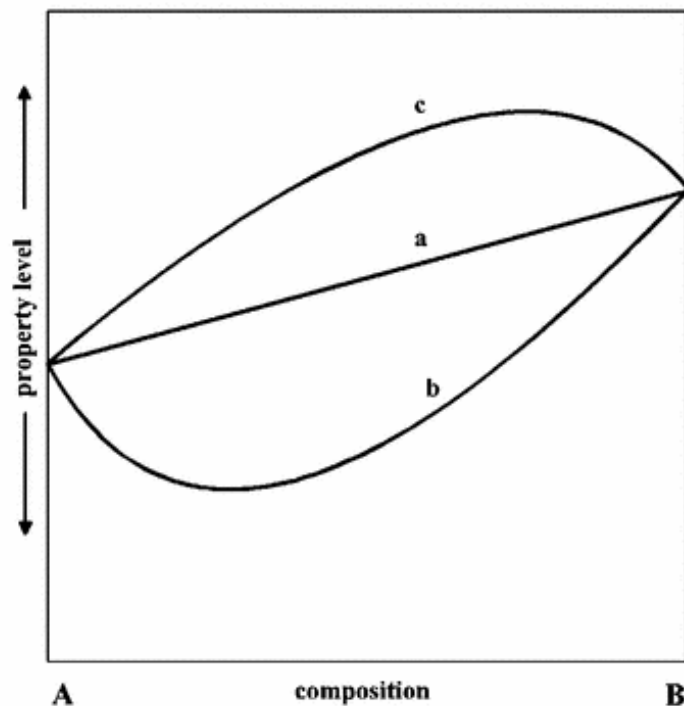
(Purely) Synthetic Block Copolymers
- Solid-State Self-Assembly -
Applications

Blend Compatibilization

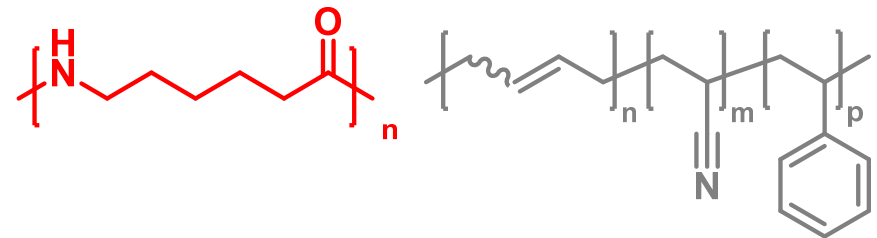
Not pure self-assembly – localization at interfaces

Why do we blend polymers?

To create materials with combinations of properties superior to the individual components. Miscibility influences greatly elastic modulus, tensile strength, glass transition temperature, viscosity...



Ex.: PA6/ABS



PA6 provides good heat and chemical resistance, tensile properties, flow, good paintability.

ABS lowers moisture absorption, improves the dimensional stability, and provides a lower price and good low temperature impact.

Influencing the Interface in Polymer Blends by Compatibilization with Block Copolymers
in *Polymer Surfaces and Interfaces*, 2008, Springer

Applications of Bulk Self-Assembly

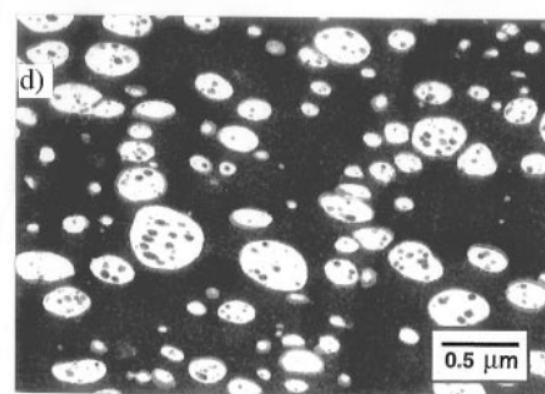
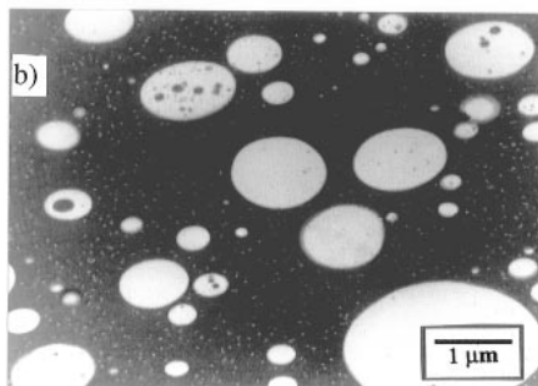
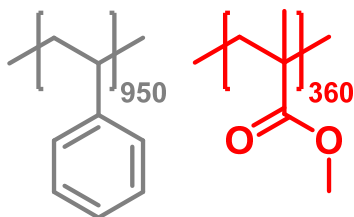
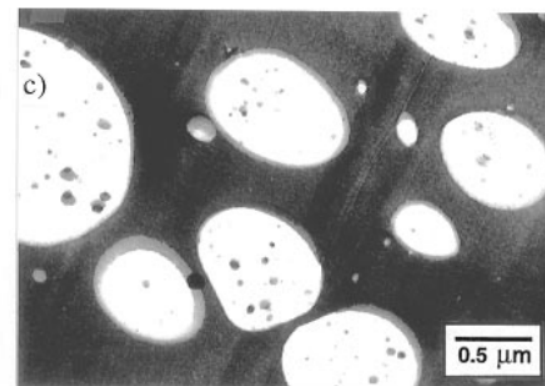
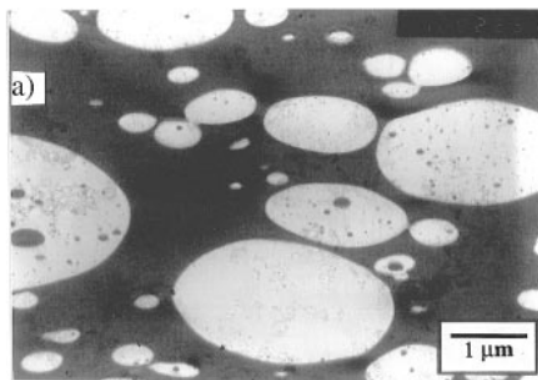
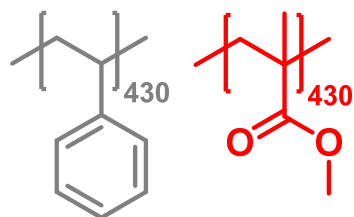
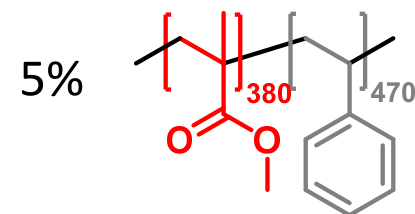
Blend Compatibilization

Not pure self-assembly – localization at interfaces

Effect: reduction of the interface

70% PS + 30% **PMMA**

No BCP



Macromolecules **1996**, 29, 5590

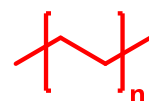
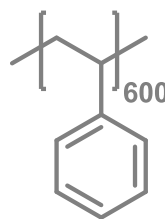
Applications of Bulk Self-Assembly

Blend Compatibilization

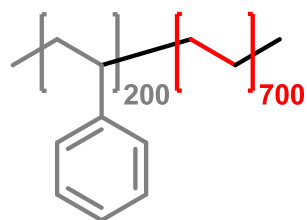
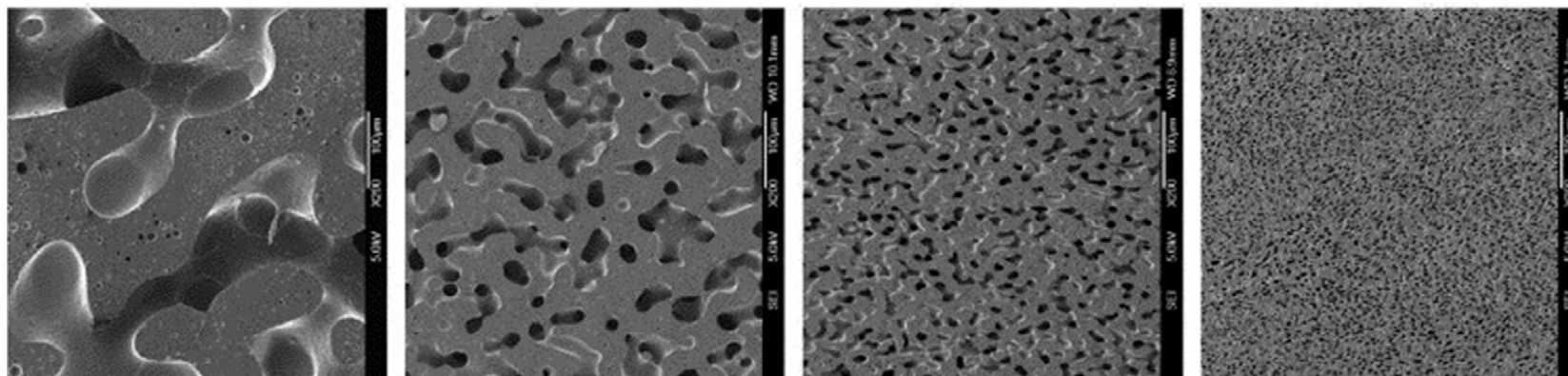
Not pure self-assembly – localization at interfaces

Effect: reduction of the interface

50% PS + 50% HDPE



$M_n = 16000 \text{ g mol}^{-1}$
 $\bar{D} = 5$



-

0.1

0.3

1

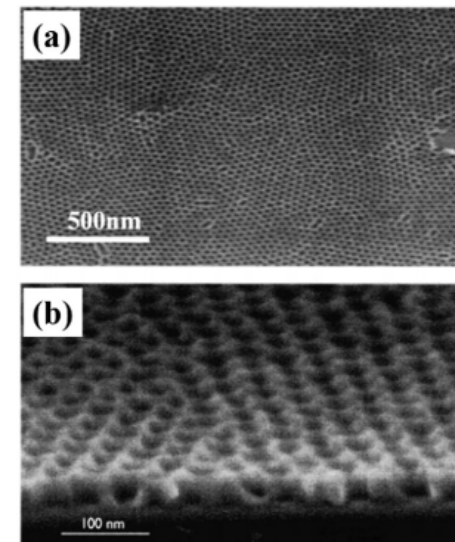
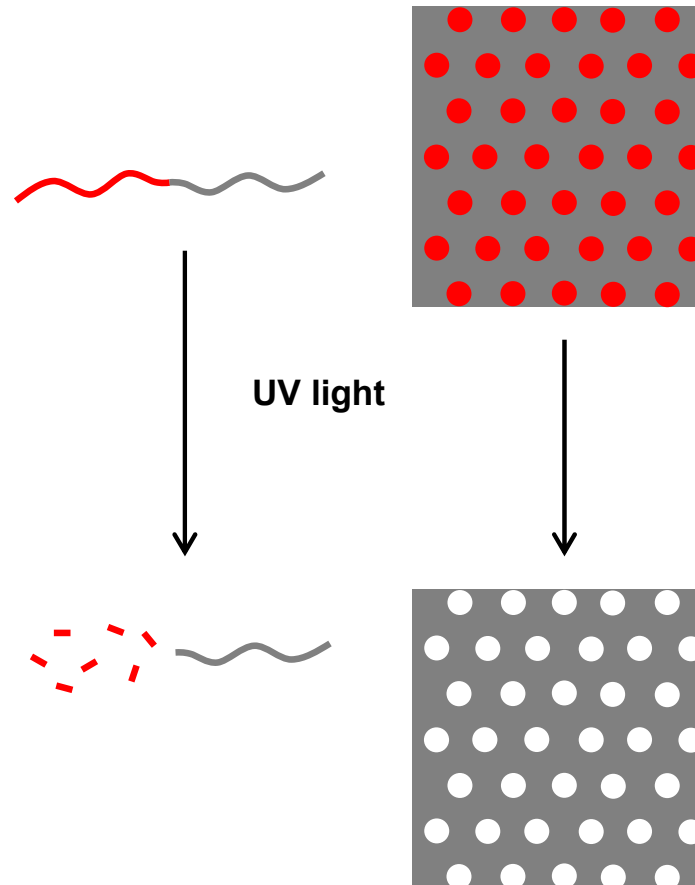
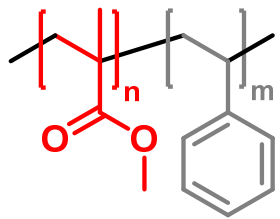
%

Polymer **2005**, 46, 183

Selective Removal of One Domain

Methods relying on the difference of degradability between the two blocks

PMMA-*b*-PS

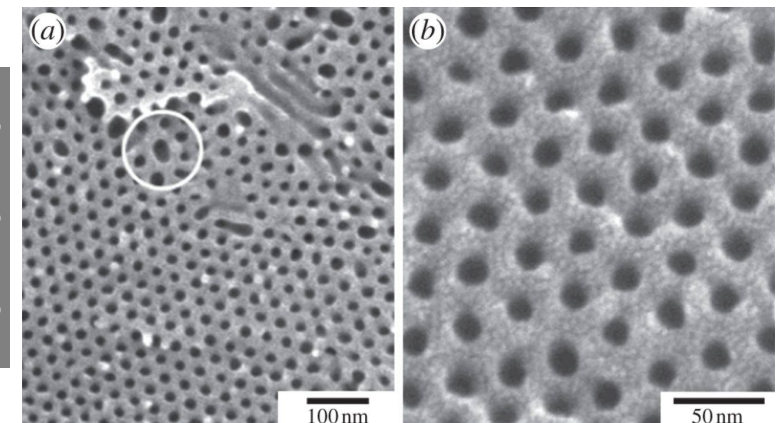
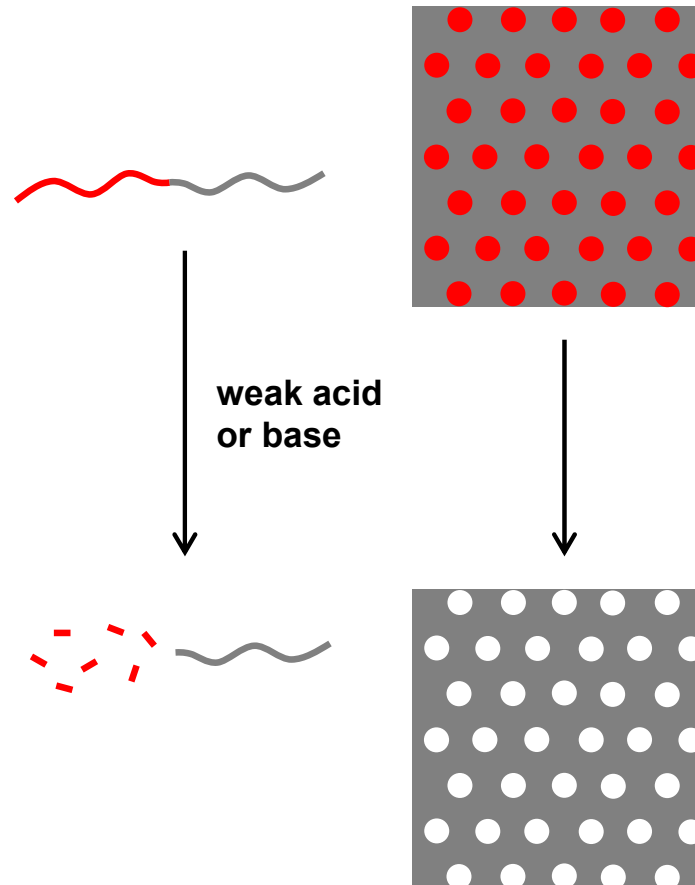
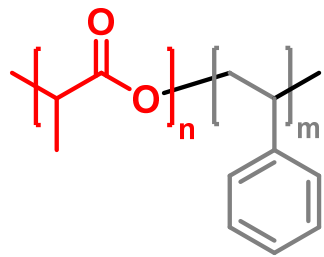


Adv. Mater. 2000, 12, 787

Selective Removal of One Domain

Methods relying on the difference of degradability between the two blocks

PLA-*b*-PS



Applications of Bulk Self-Assembly

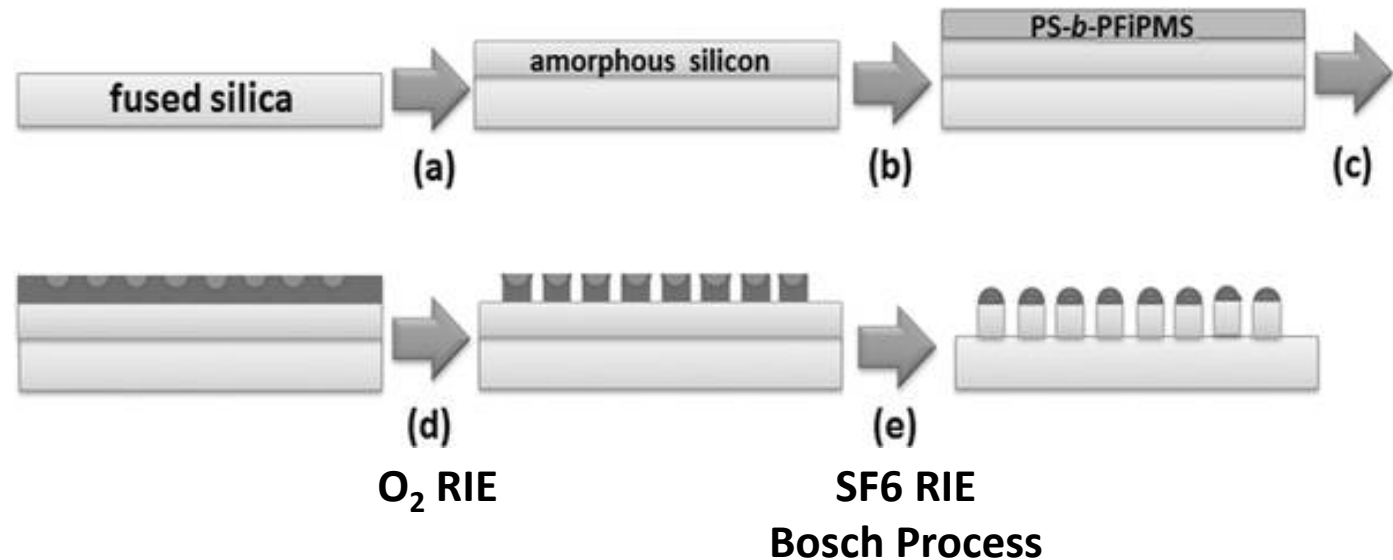
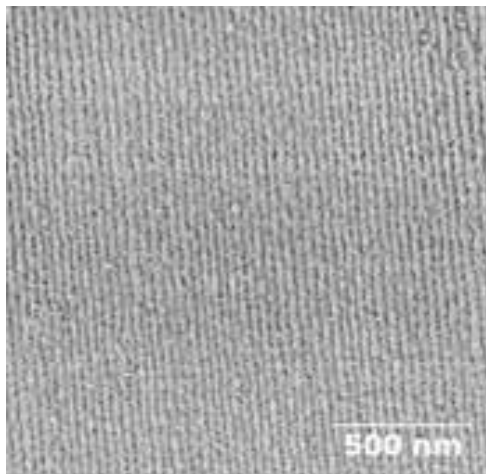
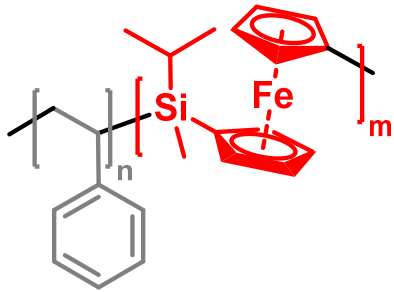
Selective Removal of One Domain

Etch Masks for Lithography – “Block Copolymer Lithography”

→ *Fabrication of Inorganic Nanomaterials*

Ex.: Wire Grid Polarizers

polystyrene-*block*-poly(ferrocenylisopropylmethylsilane)



Adv. Mater. **2014**, 26, 791

Applications of Bulk Self-Assembly

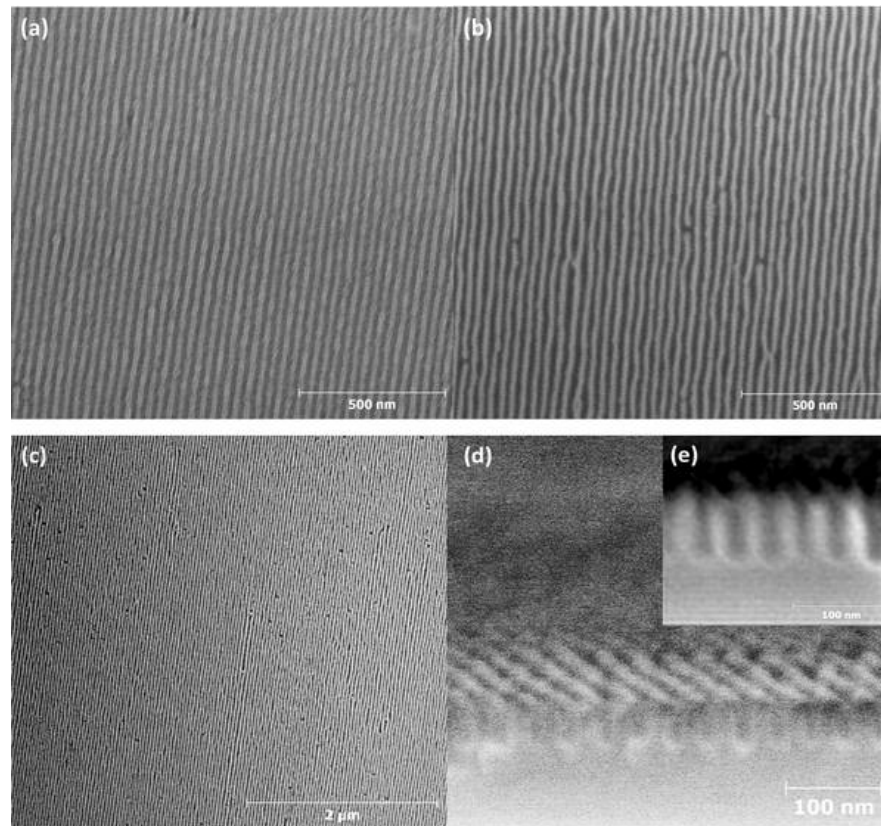
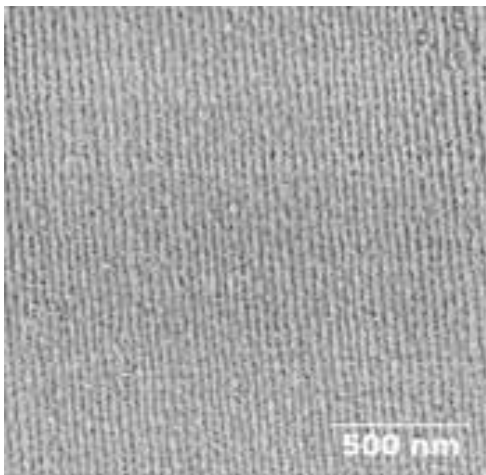
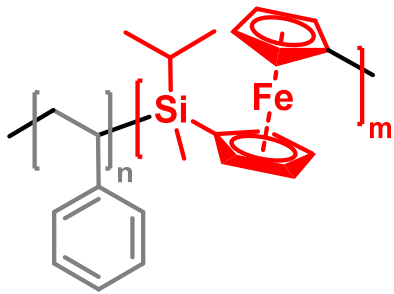
Selective Removal of One Domain

Etch Masks for Lithography – “Block Copolymer Lithography”

→ *Fabrication of Inorganic Nanomaterials*

Ex.: Wire Grid Polarizers

polystyrene-*block*-poly(ferrocenylisopropylmethylsilane)



Adv. Mater. **2014**, 26, 791

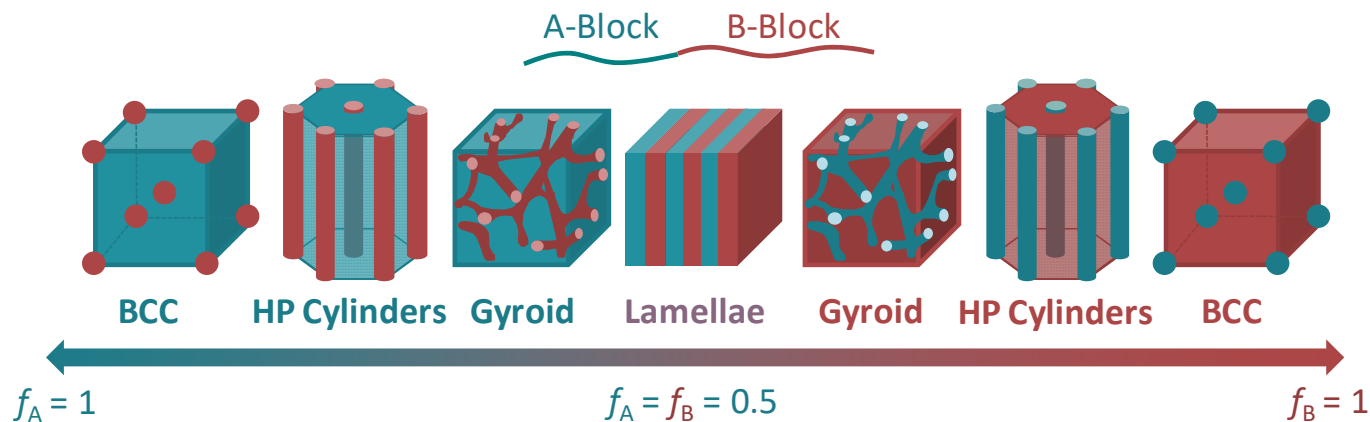
Two Important Challenges

- Low feature size (race!)
- Long range order (regularity of the patterns)

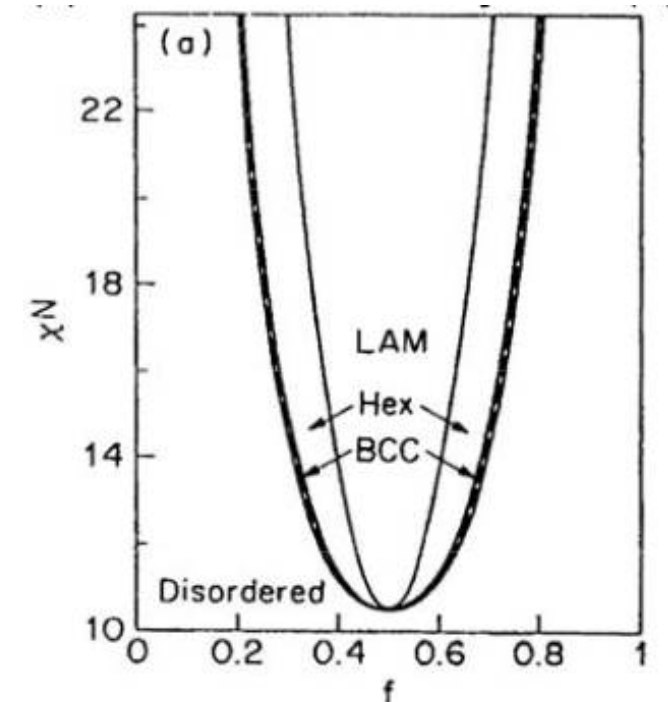
Applications of Bulk Self-Assembly

Race for lowest feature size

One needs to reduce the polymer “size”:



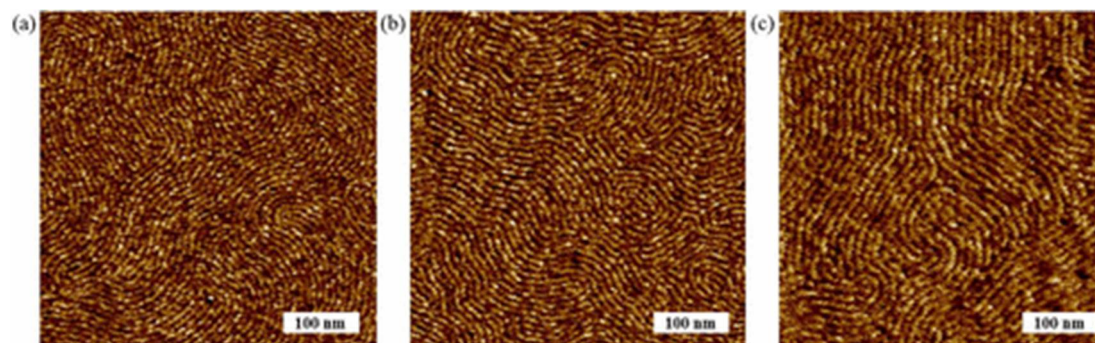
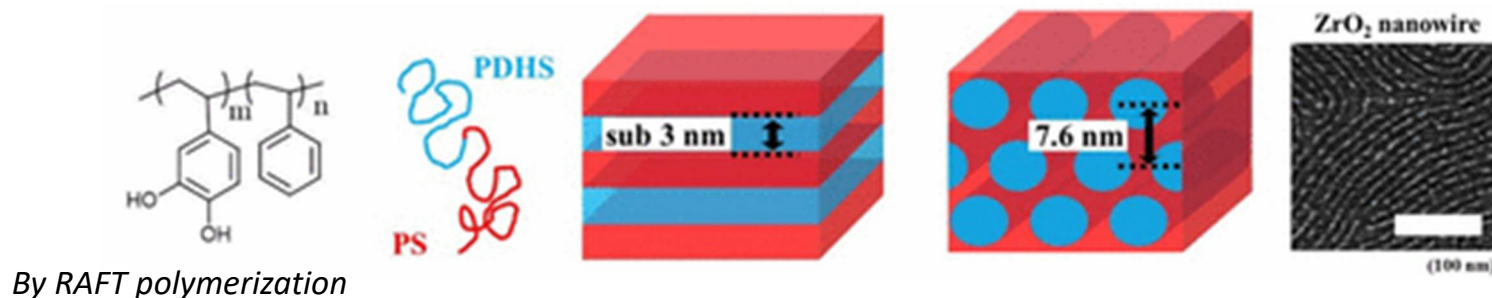
The incompatibility between the two blocks must be increased. **χ must increase!**



Applications of Bulk Self-Assembly

Race for lowest feature size

Increasing polarity difference between the blocks

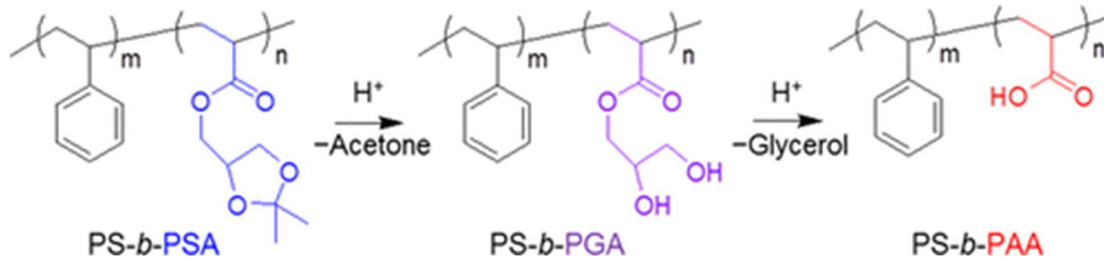


Molar mass increases

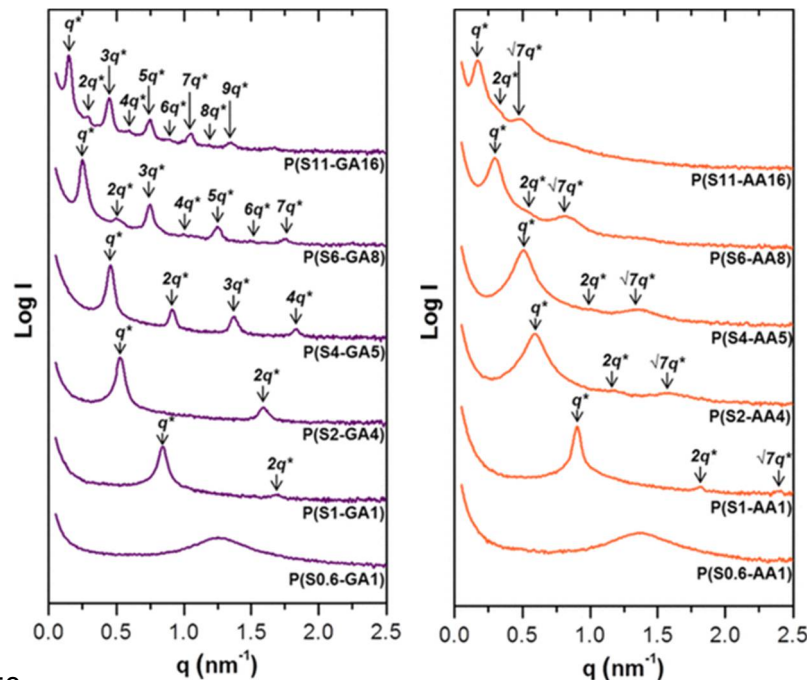
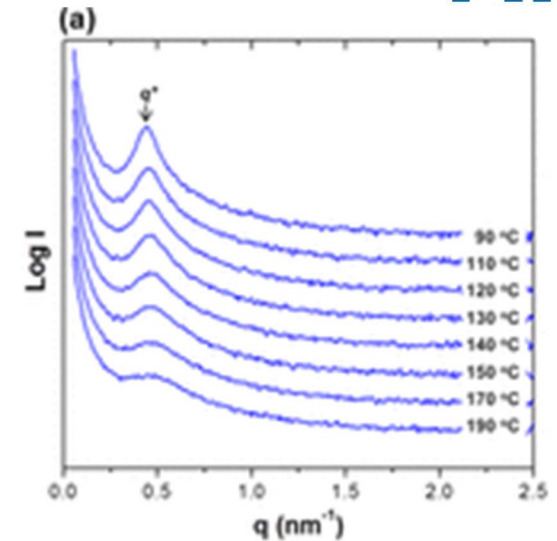
Applications of Bulk Self-Assembly

Race for lowest feature size

Increasing polarity difference between the blocks



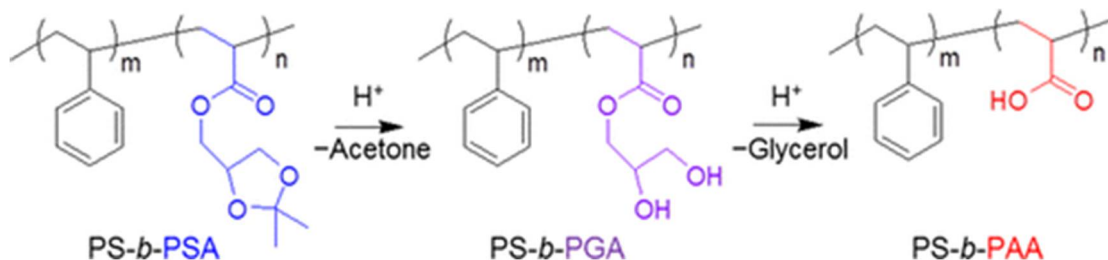
By ATRP



Macromolecules 2019, 52, 6458

Race for lowest feature size

Increasing polarity difference between the blocks

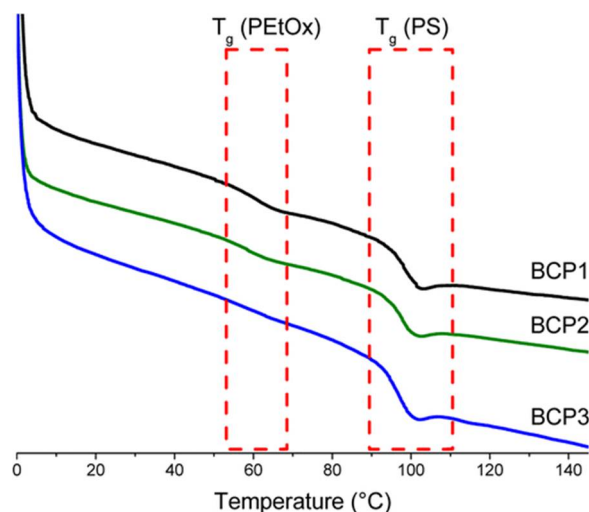
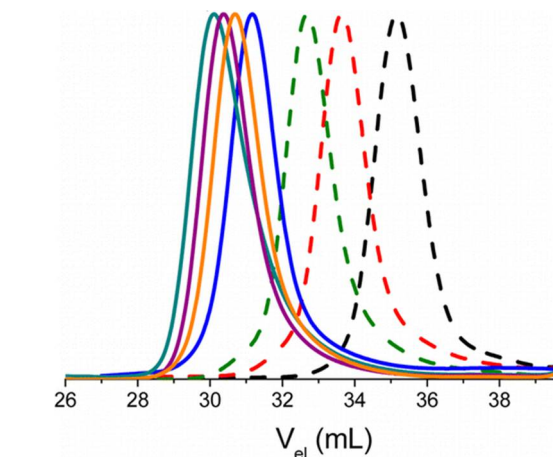


sample code	total <i>N</i>	PS- <i>b</i> -PSA <i>L</i> ₀ (nm)	PS- <i>b</i> -PGA <i>L</i> ₀ (nm)	PS- <i>b</i> -PAA <i>L</i> ₀ (nm)
P(S11-SA16)	194	23.3	42.5	36.5
P(S6-SA8)	103	14.2	25.3	20.9
P(S4-SA5)	63	disorder	13.6	12.3
P(S2-SA4)	43	disorder	11.9	10.6
P(S1-SA1)	19	disorder	7.4	6.9
P(S0.6-SA1)	13	disorder	disorder	disorder

Applications of Bulk Self-Assembly

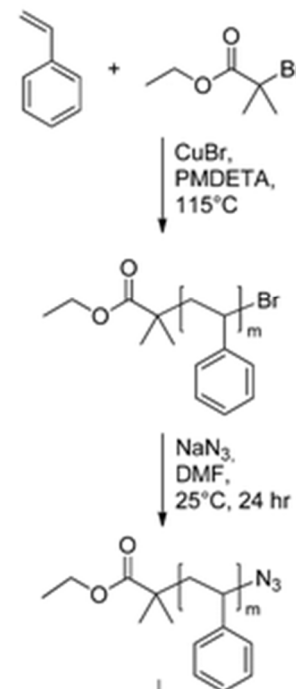
Race for lowest feature size

Using very different segments

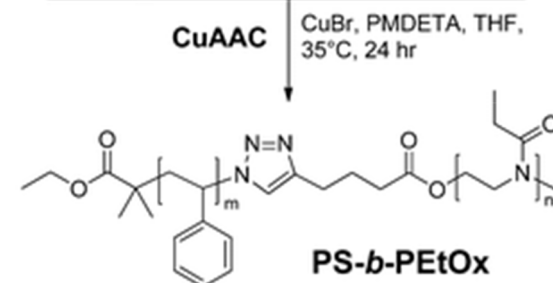
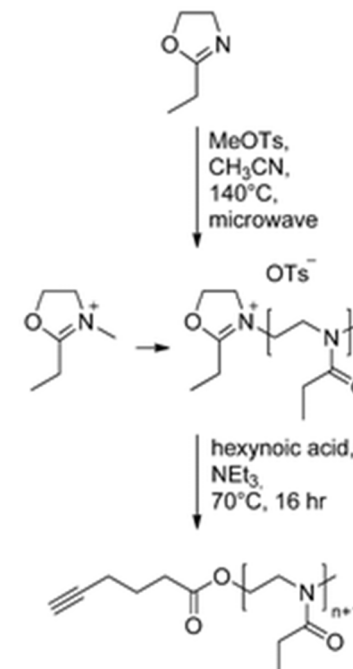


ACS Macro Lett. 2013, 2, 677

atom transfer radical
polymerization (ATRP)



cationic ring-opening
polymerization (CROP)



Race for lowest feature size

Using very different segments

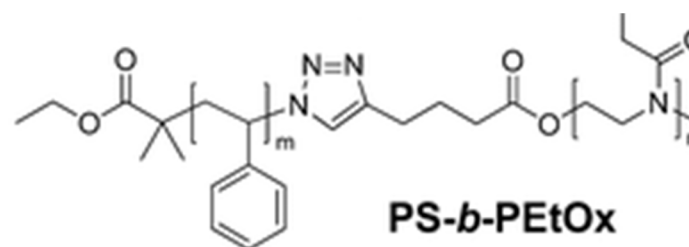
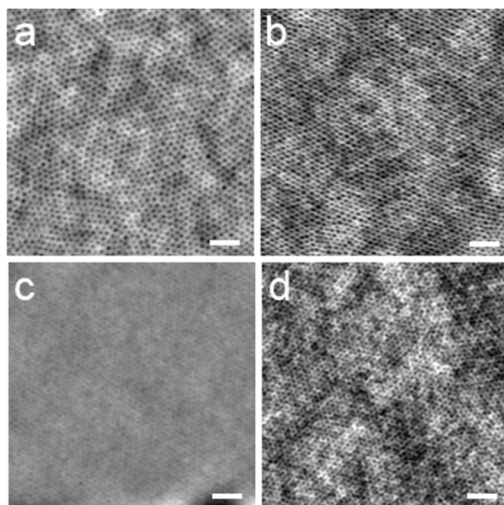


Table 2. Characteristics of the Thin Films Determined by AFM and GISAXS

sample	M_{total}^a [g mol ⁻¹]	f_{PEtOx}^b	$d\text{-spacing}^c$ [nm]	$d\text{-spacing}^d$ [nm]	feature size ^c [nm]	morphology and orientation ^{c,d}
BCP1	19000	30	22.4	21.5	13.5 ± 0.9	cylinders
BCP2	17000	22	19.6	19.1	14.6 ± 1.4	cylinders
BCP3	15000	12	18.4	19.7	12.7 ± 1.4	spheres
BCP4	12000	16	17.6	18	10.4 ± 0.9	cylinders

^aTheoretical molar mass; calculated based on the molar mass obtained for the individual building blocks. ^bVolume fraction ($\rho_{\text{PS}} = 1.05 \text{ g cm}^{-3}$, $\rho_{\text{PEtOx}} = 1.14 \text{ g cm}^{-3}$). ^cDetermined by AFM. ^dDetermined by GISAXS (all cylinders are orthogonal to the substrate).

Applications of Bulk Self-Assembly

Race for lowest feature size

Influencing the interface between the two phases

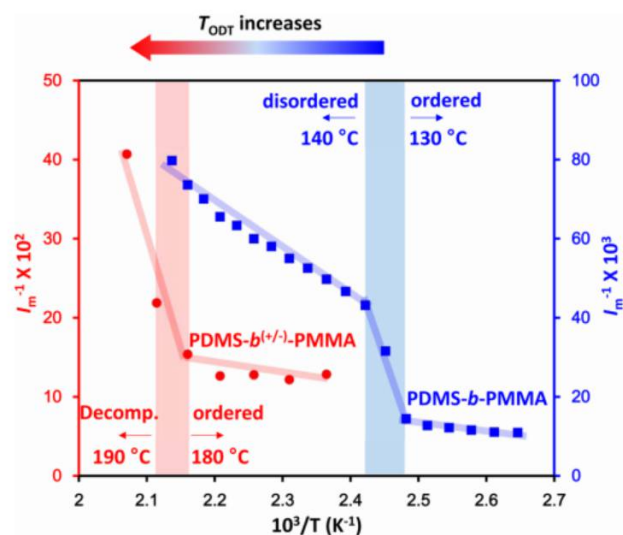
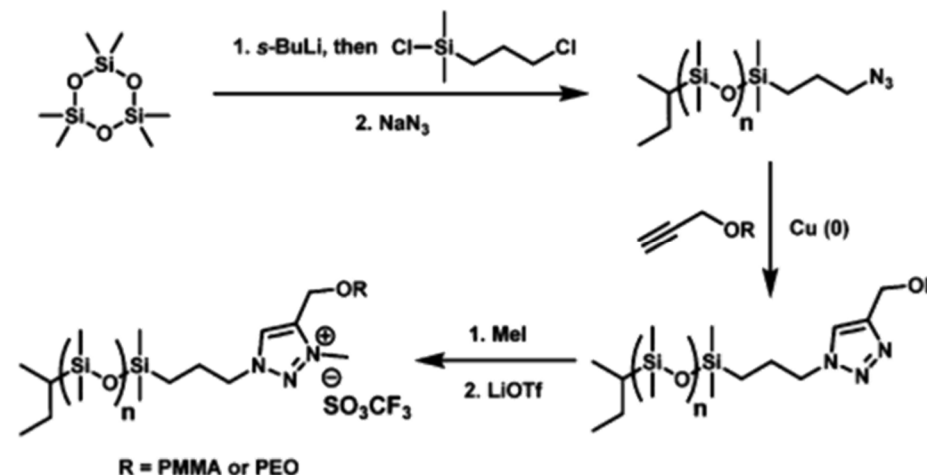
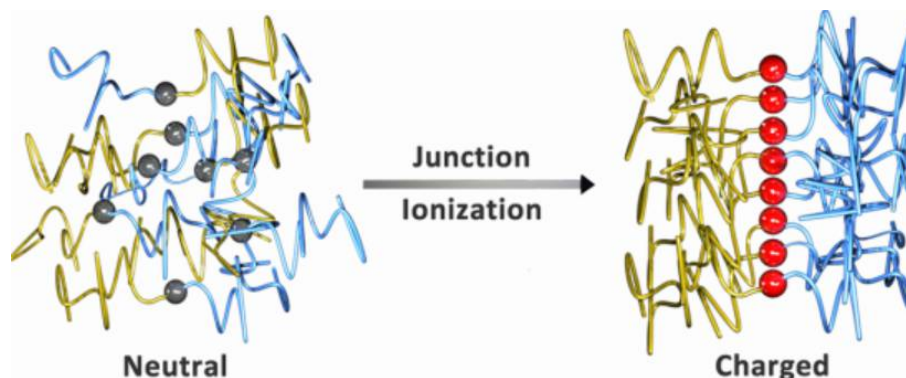
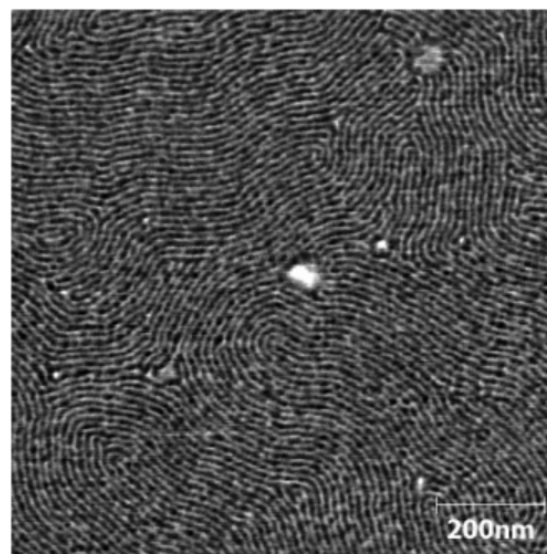


Figure 2. Characterization of order disorder transition temperatures for PDMS- b -PMMA (in blue) and PDMS- $b^{(\pm)}$ -PMMA (in red). I_m is the maximum intensity of the SAXS profile at a particular temperature.



ACS Macro Lett. 2015, 4, 1332